(Fall Semester of 2025) Final Examination of Nonlinear and Adaptive Control

January 8, 2026

P.S.: This paper was memorized and typeset after the examination ended, and there's no cheating behaviour during the examination.

1. (15 points) Answer the following questions.
   (a) Please write down the main content of this course.
   (b) Please write down the differences between indirect and direct adaptive control.

2. (10 points) Consider the system defined by the following equations:

$$\begin{aligned}\dot{x}_1 &= \frac{2}{3}x_2 \\ \dot{x}_2 &= -x_1 + x_2(1 - 3x_1^2 - 2x_2^2)\end{aligned}$$

   (a) Show that the points defined by (i) $x = (0, 0)$ and (ii) $1-(3x_1^2+2x_2^2) = 0$ are invariant sets.
   (b) Study the stability of the origin and the invariant set $1 - (3x_1^2 + 2x_2^2) = 0$, respectively, using LaSalle's Invariant Theorem.

3. (15 points) Euler equations for a rotating rigid spacecraft are given by

$$\begin{aligned}J_1\dot{\omega}_1 &= (J_2 - J_3)\,\omega_2\omega_3 + u_1, \\ J_2\dot{\omega}_2 &= (J_3 - J_1)\,\omega_3\omega_1 + u_2, \\ J_3\dot{\omega}_3 &= (J_1 - J_2)\,\omega_1\omega_2 + u_3,\end{aligned}$$

   where ω_1 to ω_3 are the components of the angular velocity vector ω along the principal axes, u_1 to u_3 are the torque inputs applied about the principal axes, and J_1 to J_3 are the principal moments of inertia.

   (a) Show that with $u_1 = u_2 = u_3 = 0$ the origin $\omega = 0$ is stable. Is it asymptotically stable?

(b) Suppose the torque inputs apply the feedback control $u_i = -k_i\omega_i$, where k_1 to k_3 are positive constants. Show that the origin of the closed-loop system is globally asymptotically stable.

4. (15 points) Consider the following nonlinear systems

$$\dot{x}_1 = -x_1^3 + \theta_1\phi_1(x_1) + x_2,$$
$$\dot{x}_2 = u + x_1,$$

where $\phi_1(x_1)$ is Lipschitz continuous and satisfying $\phi_1(0) = 0$. Design an adaptive backstepping controller $u(x_1, x_2, \hat{\theta}_1)$ such that $\lim_{t\to\infty} x_1(t) = \lim_{t\to\infty} x_2(t) = 0$.

5. (15 points) Consider a linear system with nonlinear matched uncertainties in the form

$$\dot{x} = Ax + B\left(u + \Theta^T\Phi(x)\right) + \varepsilon(t),$$

where $x \in \mathbb{R}^n$ is the state, $u \in \mathbb{R}^m$ is the control input, $A \in \mathbb{R}^{n\times n}$, $B \in \mathbb{R}^{n\times m}$, $\Lambda \in \mathbb{R}^{m\times m}$, $\Theta \in \mathbb{R}^{N\times m}$ are constant matrices, and $\varepsilon(t) \in \mathbb{R}^n$ is the disturbance. Assume that the pair $(A, B\Lambda)$ is controllable. $\Phi(x) = (\phi_1(x), \ldots, \phi_n(x))^T \in \mathbb{R}^N$ denotes the known regressor vector, whose compononts $\phi_i(x)$ are assumed to be Lipschitz-continuous in x.

(a) Assume that $\varepsilon(t) = 0$, B is known, while A and Λ are unknown. In addition, it is assumed that Λ is diagonal with m nonzero diagonal elements $\lambda_1, \lambda_2, ..., \lambda_m$, and the signs of all λ_i are known. Design and analyze a directed MRAC scheme that can stablize the system and regulate x towards zero.

(b) Assume that $\|\varepsilon(t)\| \le \varepsilon_f, \forall t > 0$, where $\varepsilon_f > 0$, and A, B, Λ are all known, while Λ is a positive diagonal matrix. Design a σ-modification robust control algorithm that can stabilize the system and regulate x towards the neighborhood of zero.

Hint:

$$\lambda_{\min}(P)x^Tx \le x^TPx \le \lambda_{\max}(P)x^Tx, \quad \forall P = P^T > 0$$
$$x^TP\varepsilon \le \|x\|\|P\|\|\varepsilon\|$$
$$\frac{1}{\lambda_{\max}(\Gamma)}\operatorname{tr}(\Theta\Theta^T) \le \operatorname{tr}(\Theta\Gamma^{-1}\Theta^T) \le \frac{1}{\lambda_{\min}(\Gamma)}\operatorname{tr}(\Theta\Theta^T)$$
$$2\operatorname{tr}(\Theta\tilde{\Theta}^T) \le \operatorname{tr}(\Theta\Theta^T) + \operatorname{tr}(\tilde{\Theta}\tilde{\Theta}^T)$$

6. (30 points) The nonlinear dynamic equations for an m-link robot take the form

$$M(q)\ddot{q} + C(q,\dot{q})\dot{q} + D\dot{q} + g(q) + d(q,\dot{q},t) = u,$$

where $q \in \mathbb{R}^p$ is the vector of generalized coordinates representing the joint positions, $M(q) \in \mathbb{R}^{p \times p}$ is the symmetric positive-definite inertia matrix, $C(q, \dot{q})\dot{q} \in \mathbb{R}^p$ is the vector of Coriolis and centrifugal torques, $D\dot{q}$ is the vector of viscous damping, $g(q) = [\partial P(q)/\partial q]^T \in \mathbb{R}^p$ is the gravitational torque, $P(q)$ is a positive definite function of q representing the total potential energy of the links due to gravity, and $u \in \mathbb{R}^p$ is the control torque. The following assumptions hold:

(A1) There exist positive constants $k_{\underline{m}}$ and $k_{\overline{m}}$ such that $0 < k_{\underline{m}} I_p \leq M(q) \leq k_{\overline{m}} I_p$. For $x, y, z \in \mathbb{R}^p$, $0 \leq \|C(x, y)z\| \leq K_C \|y\| \|z\|$.

(A2) $\dot{M}(q) - 2\dot{C}(q, \dot{q})$ is skew symmetric and D is positive semidefinite.

(A3) $g(q) = 0$ has an isolated root at $q = 0$.

(A4) There are parameter uncertainties in $M(q), C(q, \dot{q})$, D and $g(q)$. For $x, y, z \in \mathbb{R}^p$, $M(q)x + C(q, \dot{q})y + Dz + g(q) = Y(q, \dot{q}, x, y, z)\Theta$, with Θ being an unknown constant vector.

(a) Assume that $d(q, \dot{q}, t) = 0$. With $u = 0$, use the total energy $V(q, \dot{q}) = \frac{1}{2}\dot{q}^T M(q)\dot{q} + P(q)$ as a Lyapunov function candidate to show that the origin ($q = 0, \dot{q} = 0$) is stable.

(b) Assume that $d(q, \dot{q}, t) = 0$. Design an adaptive controller (using sliding mode variables) such that $q(t)$ asymptotically tracks a reference trajectory $q_d(t)$, where $q_d(t)$, $\dot{q}_d(t)$, and $\ddot{q}_d(t)$ are continuous and bounded.

(c) Assume that $d(q, \dot{q}, t) = d_1(t) + d_2(t)q + d_3(t)\dot{q}$, where $d_1(t), d_2(t), d_3(t)$ are time-varying continuous functions satisfying

$$\max_{t \geq 0} \{\|d_1(t)\|, \|d_2(t)\|, \|d_3(t)\|\} \leq d_{\max},$$

for some unknown positive constant $d_{\max}$. Design an adaptive controller (using sliding mode variables) such that $q(t)$ asymptotically tracks a reference trajectory $q_d(t)$, where $q_d(t)$, $\dot{q}_d(t)$, and $\ddot{q}_d(t)$ are continuous and bounded.